Practical no. 14

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Program :-

```
#include<stdio.h>
#include<stdlib.h>
#include<conio.h>
struct Node{
int coefficient;
int exponent;
struct Node*next;
};
struct Node*createNode(int coeff,int exp){
struct Node*newNode=(struct Node*)malloc(sizeof(struct Node));
newNode->coefficient=coeff;
newNode->exponent=exp;
newNode->next=NULL;
return newNode;
}
struct Node*insertTerm(struct Node*head,int coeff,int exp){
struct Node*current=head;
struct Node*newNode=createNode(coeff,exp);
if(head==NULL||exp>head->exponent){
newNode->next=head;
return newNode;
}
while(current->next!=NULL&&exp<current->next->exponent){
```

```c
current=current->next;
}
newNode->next=current->next;
current->next=newNode;
return head;
}
void display(struct Node*head){
struct Node*current=head;
if(head==NULL){
printf("\nList is empty!!");
}
while(current!=NULL){
printf("%dx^%d",current->coefficient,current->exponent);
current=current->next;
if(current!=NULL){
printf("+");
}
}
printf("\n");
}
void main(){
struct Node*poly1=NULL;
struct Node*poly2=NULL;
struct Node*head;
poly1=insertTerm(poly1,5,3);
poly1=insertTerm(poly1,7,2);
poly1=insertTerm(poly1,1,0);
```

```
poly2=insertTerm(poly2,4,2);

poly2=insertTerm(poly2,2,1);

poly2=insertTerm(poly2,1,0);

clrscr();

printf("\nPolynomial 1:");

display(poly1);

printf("\nPolynomial 2:");

display(poly2);

getch();

}
```

<u>OUTPUT</u> :-